MULTIMEDIA TECHNOLOGY

Unit-3: Graphics in Multimedia

Comprehensive Study Notes

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1. Importance of Graphics in Multimedia

Definition

Graphics are visual representations of information that include images, drawings, charts, photographs, and other visual elements. In multimedia, graphics serve as the primary visual medium that captures attention and conveys information effectively.

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+-------------------------------------------------------------------+
|            IMPORTANCE OF GRAPHICS IN MULTIMEDIA                    |
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Why Graphics are Essential in Multimedia

Reason	Explanation	Example
Attention Grabbing	Visuals capture attention faster than text	Website hero images
Clarity	Explains concepts more clearly	Infographics, diagrams
Emotional Impact	Evokes emotions and feelings	Photography, art
Universal Communication	Transcends language barriers	Traffic signs, icons
Memory Aid	Improves information retention	Educational visuals
Professional Appearance	Creates polished presentation	Corporate design
Navigation Aid	Guides users through content	Menu icons, buttons
Storytelling	Enhances narrative delivery	Comic strips, storyboards

2. Vector and Raster Graphics

Raster Graphics (Bitmap Graphics)

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|                       RASTER GRAPHICS                           |
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|                                                                  |
|   +-------------------------------------------------------+     |
|   |                                                       |     |
|   | DEFINITION: Image made up of pixels arranged in a grid |     |
|   | Also Called: Bitmap images                            |     |
|   | Resolution Dependence: Yes (pixel-based)              |     |
|   |                                                       |     |
|   +-------------------------------------------------------+     |
|                                                                  |
|   CHARACTERISTICS:                                                |
|   * Made of pixels                                               |
|   * Loses quality when scaled up (pixelation)                   |
|   * Fixed resolution                                             |
|   * File size depends on pixel count                             |
|   * Can show rich color variations                               |
|   * Suitable for photographs and complex images                  |
|   * Each pixel stores color information                          |
|                                                                  |
|   FILE FORMATS: JPEG, PNG, GIF, BMP, TIFF                       |
|                                                                  |
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Vector Graphics

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|                       VECTOR GRAPHICS                           |
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|                                                                  |
|   +-------------------------------------------------------+     |
|   |                                                       |     |
|   | DEFINITION: Image made of mathematical paths/objects   |     |
|   | Also Called: Object-oriented graphics                 |     |
|   | Resolution Dependence: No (mathematical)              |     |
|   |                                                       |     |
|   +-------------------------------------------------------+     |
|                                                                  |
|   CHARACTERISTICS:                                                |
|   * Made of mathematical objects (paths, points)                 |
|   * Infinite scalability (no quality loss)                       |
|   * Resolution independent                                       |
|   * Smaller file size (for simple graphics)                      |
|   * Perfect for logos, icons, diagrams                           |
|   * Supports transparency                                        |
|   * Editable at object level                                     |
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|                                                                   |
|   FILE FORMATS: EPS, SVG, AI, CDR, WMF                           |
|                                                                   |
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Vector vs Raster Comparison

Feature	Vector Graphics	Raster Graphics
Composition	Mathematical paths	Pixels
Resolution	Resolution independent	Resolution dependent
Scalability	Infinite (no quality loss)	Limited (pixelates)
File Size	Smaller (simple images)	Larger (depends on pixels)
Color Range	Limited per object	Unlimited per pixel
Photographic	Not suitable	Suitable
Editing	Object-based	Pixel-based
Compression	Not needed	Various compression
Best For	Logos, icons, diagrams	Photos, realistic images
Transparency	Excellent support	Alpha channel support
Software	Illustrator, CorelDRAW, Inkscape	Photoshop, GIMP, Paint

Scalability Comparison

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+-----------------------------------------------------------------+
|                    SCALABILITY COMPARISON                         |
+-----------------------------------------------------------------+
|                                                                   |
|   RASTER IMAGE (Pixel-based):                                     |
|   Original Size:       Scaled Up:        Scaled More:             |
|   +----------+       +----------+       +-------------+           |
|   | +----+  |       | +----+   |       | +--------+  |          |
|   | |    |  |       | |    |   |       | |        |  |          |
|   | |    |  |       | |    |   |       | |        |  |          |
|   | +----+  |       | +----+   |       | +--------+  |          |
|   +----------+       +----------+       +-------------+           |
|   Result: Pixelated/Blurry                                        |
|                                                                   |
|   VECTOR IMAGE (Mathematical):                                    |
```

```
|   Original Size:       Scaled Up:        Scaled More:                  |
|   +----------+       +----------+       +-------------+                |
|   | +----+  |       | +----+    |       | +--------+  |               |
|   | |    |  |       | |    |    |       | |        |  |               |
|   | +----+  |       | +----+    |       | +--------+  |               |
|   +----------+       +----------+       +-------------+                |
|   Result: Always Sharp and Clear (No Quality Loss)                     |
|                                                                        |
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3. Image Capturing Methods

3.1 Scanner

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+-----------------------------------------------------------------+
|                      SCANNER TECHNOLOGY                         |
+-----------------------------------------------------------------+
|                                                                 |
|   WORKING PRINCIPLE:                                            |
|   +----------+  +----------+  +----------+                     |
|   | Document|--->|   CCD    |--->| Digital |                    |
|   | (Light   |  |  Sensor  |  |  Image   |                     |
|   |  Source) |  |          |  |          |                     |
|   +----------+  +----------+  +----------+                     |
|                                                                 |
|   TYPES OF SCANNERS:                                            |
|                                                                 |
|   1. FLATBED SCANNER                                            |
|      - Most common type                                         |
|      - Place document on glass plate                            |
|      - Good for photos, documents, books                        |
|      - Resolution: 300-4800 dpi                                 |
|                                                                 |
|   2. SHEETFED SCANNER                                           |
|      - Feeds paper through rollers                              |
|      - Faster for multiple pages                                |
|      - Business document scanning                               |
|      - Resolution: 300-600 dpi                                  |
|                                                                 |
|   3. DRUM SCANNER                                               |
|      - Professional grade                                       |
|      - Highest quality                                          |
|      - Uses photomultiplier tubes                               |
|      - Resolution: Up to 8000 dpi                               |
|      - Very expensive                                           |
|                                                                 |
|   4. HANDHELD SCANNER                                           |
|      - Portable                                                 |
|      - Manual operation                                         |
|      - Lower quality                                            |
|      - Resolution: 100-300 dpi                                  |
|                                                                 |
|   SCANNER PARAMETERS:                                           |
|   * Resolution (dpi) - Quality level                            |
|   * Color Depth - Number of colors captured                     |
|   * Scan Area - Maximum document size                           |
|   * Bit Depth - 24-bit, 48-bit color                            |
|                                                                 |
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3.2 Digital Camera

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|                    DIGITAL CAMERA TECHNOLOGY                            |
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|                                                                         |
|   WORKING PRINCIPLE:                                                    |
|   +----------+  +----------+  +----------+                             |
|   |  Lens    |--->|  Image   |--->| Digital |                                |
|   | (Focus)  |  |  Sensor  |  | Memory  |                              |
|   |          |  | (CCD/    |  | (Image  |                              |
|   |          |  |  CMOS)   |  | Storage)|                              |
|   +----------+  +----------+  +----------+                             |
|                                                                         |
|   IMAGE SENSOR TYPES:                                                   |
|                                                                         |
|   CCD (Charge-Coupled Device):                                          |
|   * Higher image quality                                                |
|   * Better low-light performance                                        |
|   * More expensive                                                      |
|   * Used in professional cameras                                        |
|   * Higher power consumption                                            |
|                                                                         |
|   CMOS (Complementary Metal-Oxide-Semiconductor):                       |
|   * Lower power consumption                                             |
|   * Lower cost                                                          |
|   * Faster processing                                                   |
|   * Good quality (improving)                                            |
|   * Used in consumer cameras, phones                                    |
|                                                                         |
|   KEY CAMERA SPECIFICATIONS:                                            |
|   * Megapixels (MP) --> Image Resolution                                |
|   * Sensor Size --> Light Capture Ability                               |
|   * Aperture (f/stop) --> Depth of Field                               |
|   * Shutter Speed --> Motion Capture Ability                            |
|   * ISO Speed --> Light Sensitivity                                     |
|   * Zoom --> Focal Length Range                                         |
|                                                                         |
|   DIGITAL CAMERA TYPES:                                                 |
|   * Compact/Point & Shoot - Consumer use                               |
|   * DSLR (Digital Single Lens Reflex) - Professional                   |
|   * Mirrorless - Modern professional                                    |
|   * Smartphone - Everyday use                                           |
|   * Action Camera - Sports/POV                                          |
|                                                                         |
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```

3.3 Other Image Capturing Methods

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|                 OTHER IMAGE CAPTURING METHODS                           |
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|                                                                         |
|   SCREEN CAPTURE (Screenshot):                                          |
|   * Capture digital display content                                     |
|   * Software-based method                                               |
|   * Used for tutorials, documentation                                   |
|   * Instant capture of screen activity                                  |
|   * Can capture full screen or region                                   |
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|                                                                    |
|   GRAPHICS TABLET:                                                 |
|   * Creates digital art manually                                   |
|   * Uses pressure-sensitive pen                                    |
|   * Used by artists and designers                                  |
|   * Software-based creation                                        |
|   * Requires artistic skills                                       |
|                                                                    |
|   3D SCANNING:                                                     |
|   * Creates 3D models of real objects                              |
|   * Used for 3D printing, AR/VR                                    |
|   * Uses laser/LIDAR technology                                    |
|   * Captures shape, texture, color                                 |
|   * Emerging technology                                            |
|                                                                    |
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```

4. Various Attributes of Images

4.1 Image Size

```
+------------------------------------------------------------------+
|                    IMAGE SIZE ATTRIBUTES                         |
+------------------------------------------------------------------+
|                                                                  |
|   IMAGE SIZE MEASUREMENTS:                                       |
|                                                                  |
|   1. DIMENSIONS (Pixel Size)                                     |
|      Width x Height in pixels                                    |
|      Example: 1920 x 1080 pixels                                 |
|      Determines level of detail                                  |
|      Affects file size                                           |
|                                                                  |
|   2. PRINT SIZE (Physical Size)                                  |
|      Width x Height in inches/cm                                 |
|      Example: 10" x 8"                                           |
|      Depends on DPI/PPI                                          |
|      Used for printing purposes                                  |
|                                                                  |
|   3. RESOLUTION (DPI/PPI)                                        |
|      DPI = Dots Per Inch (Printing)                              |
|      PPI = Pixels Per Inch (Screen)                              |
|      Higher DPI = Better quality                                 |
|      Common: 72 PPI (Web), 300 DPI (Print)                       |
|                                                                  |
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```

Image Size Calculation

File Size (bytes) = Width x Height x Bit Depth / 8

Example: 1920 x 1080 x 24-bit / 8 = 6.22 MB

Resolution	Dimensions	Use Case
480p (SD)	640x480	Standard definition
720p (HD)	1280x720	High definition
1080p (Full HD)	1920x1080	Full HD
2K (QHD)	2560x1440	Quad HD
4K (Ultra HD)	3840x2160	Ultra HD

8K	7680x4320	Ultra HD 8K

4.2 Color Depth (Bit Depth)

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|                    COLOR DEPTH / BIT DEPTH                      |
+----------------------------------------------------------------+
|                                                                 |
|   DEFINITION: Number of bits used to represent color of each pixel  |
|                                                                 |
|   1-bit (Monochrome)      2 colors     2^1 = 2 colors          |
|   4-bit (16 colors)      16 colors     2^4 = 16 colors         |
|   8-bit (256 colors)     256 colors    2^8 = 256 colors        |
|   16-bit (High Color)    65,536 colors 2^16 = 65,536 colors    |
|   24-bit (True Color)    16.7M colors  2^24 = 16.7M colors     |
|   32-bit (True Color+)   16.7M + alpha 2^32 = 4.29B colors     |
|                                                                 |
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```

Bit Depth	Colors	Use Case
1-bit	2 colors	Line art, sketches
4-bit	16 colors	Basic graphics
8-bit	256 colors	GIF, early web images
16-bit	65,536	Display devices
24-bit	16.7M	Photographs, web images
32-bit	16.7M + alpha	Images with transparency

4.3 Image Resolution

```
+----------------------------------------------------------------+
|                      IMAGE RESOLUTION                          |
+----------------------------------------------------------------+
|                                                                 |
|   DEFINITION: The level of detail in an image measured in DPI/PPI  |
|                                                                 |
|   TYPES OF RESOLUTION:                                         |
|                                                                 |
|   1. PPI (Pixels Per Inch) - Screen Resolution                 |
|      - Used for digital displays                               |
|      - Affects display quality                                  |
|      - Common: 72 PPI, 96 PPI                                  |
|      - Higher PPI = Sharper display                             |
|                                                                 |
```

```
|   2. DPI (Dots Per Inch) - Print Resolution                          |
|      - Used for printers                                             |
|      - Affects print quality                                         |
|      - Common: 300 DPI (standard), 600 DPI (high)                   |
|      - Higher DPI = Better print quality                             |
|                                                                      |
|   3. Screen Resolution (Display Resolution)                          |
|      - Total pixel count (Width x Height)                            |
|      - Example: 1920 x 1080 pixels                                   |
|      - Determines display capability                                 |
|                                                                      |
|   4. Image Resolution (Pixel Count)                                  |
|      - Total number of pixels in image                               |
|      - Megapixels = (Width x Height) / 1,000,000                    |
|      - Example: 12 MP = 4000 x 3000 pixels                           |
|                                                                      |
|   5. Optical Resolution (Scanner/Camera)                             |
|      - Maximum detail capture ability                                |
|      - Determined by hardware                                        |
|      - Scanner: 4800 dpi optical                                     |
|      - Camera: Based on sensor megapixels                            |
|                                                                      |
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Purpose	DPI/PPI	File Size (approx)
Web Display	72 PPI	Small (optimized)
Email	72 PPI	Small (fast loading)
Photo Print	300 DPI	Large
Magazine	300 DPI	Large
Billboard	30 DPI	Very Large
Newspaper	150 DPI	Medium
Poster	200 DPI	Large

4.4 Image Attributes Summary

```
+----------------------------------------------------------------------+
|                    COMPLETE IMAGE ATTRIBUTES                         |
+----------------------------------------------------------------------+
|                                                                      |
|   1. Size (Dimensions)                                               |
|      - Width x Height in pixels                                      |
|      - Affects detail and file size                                  |
|                                                                      |
|   2. Color Depth (Bit Depth)                                         |
|      - Number of bits per pixel                                      |
|      - Determines number of colors                                   |
```

```
|                                                                         |
|   3. Resolution (DPI/PPI)                                               |
|      - Pixels per inch (digital)                                        |
|      - Dots per inch (print)                                            |
|                                                                         |
|   4. File Format                                                        |
|      - Determines compression method                                    |
|      - Affects compatibility                                            |
|                                                                         |
|   5. Compression                                                        |
|      - Lossy (JPEG) vs Lossless (PNG)                                   |
|      - Affects file size and quality                                    |
|                                                                         |
|   6. Aspect Ratio                                                       |
|      - Width to height ratio                                            |
|      - Common: 4:3, 16:9, 3:2                                           |
|                                                                         |
|   7. Color Space/Profile                                                |
|      - RGB (Red, Green, Blue) - Digital                                 |
|      - CMYK (Cyan, Magenta, Yellow, Black) - Print                      |
|      - Grayscale - Black and white                                      |
|                                                                         |
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```

5. Various Image File Formats

5.1 BMP (Bitmap)

```
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|                        BMP FILE FORMAT                           |
+------------------------------------------------------------------+
|                                                                  |
|   FULL NAME: Bitmap File Format                                  |
|   DEVELOPED BY: Microsoft                                        |
|   FILE EXTENSION: .bmp                                           |
|   TYPE: Uncompressed / Lossless                                  |
|                                                                  |
|   CHARACTERISTICS:                                               |
|   * Windows native format                                        |
|   * Supports various bit depths (1, 4, 8, 16, 24, 32)           |
|   * Uncompressed (large file size)                               |
|   * RLE compression available (simple)                           |
|   * Simple to implement                                          |
|   * No metadata support                                          |
|   * Limited platform support                                     |
|                                                                  |
|   ADVANTAGES:                                                    |
|   * High quality (no compression loss)                           |
|   * Simple structure                                             |
|   * Wide Windows support                                         |
|                                                                  |
|   DISADVANTAGES:                                                 |
|   * Very large file size                                         |
|   * No transparency support                                      |
|   * Not suitable for web                                         |
|   * Limited metadata                                             |
|                                                                  |
|   USE CASE: Windows graphics, icon storage                       |
|                                                                  |
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```

5.2 DIB (Device Independent Bitmap)

```
+------------------------------------------------------------------+
|                   DIB FILE FORMAT                                |
+------------------------------------------------------------------+
|                                                                  |
|   FULL NAME: Device Independent Bitmap                           |
|   DEVELOPED BY: Microsoft                                        |
|   FILE EXTENSION: .dib                                           |
|   TYPE: Uncompressed                                             |
|                                                                  |
|   CHARACTERISTICS:                                               |
|   * Similar to BMP                                               |
|   * Color information independent of device                      |
```

```
|   * Contains color palette                                        |
|   * Can be used with different display devices                    |
|   * Uncompressed format                                           |
|   * Used in Windows applications                                 |
|                                                                   |
|   DIFFERENCE FROM BMP:                                            |
|   * DIB stores color data in device-independent form              |
|   * Better color consistency across devices                       |
|   * Contains additional color space information                   |
|                                                                   |
|   USE CASE: Application resources, clipboard operations           |
|                                                                   |
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```

5.3 EPS (Encapsulated PostScript)

```
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|                    EPS FILE FORMAT                              |
+----------------------------------------------------------------+
|                                                                 |
|   FULL NAME: Encapsulated PostScript                           |
|   DEVELOPED BY: Adobe Systems                                  |
|   FILE EXTENSION: .eps                                         |
|   TYPE: Vector (with possible raster)                          |
|                                                                 |
|   CHARACTERISTICS:                                              |
|   * Vector-based format                                         |
|   * Can include both vector and raster                          |
|   * PostScript language based                                   |
|   * Device independent                                          |
|   * Supports transparency                                       |
|   * Can include embedded preview image                          |
|                                                                 |
|   ADVANTAGES:                                                   |
|   * Scalable (no quality loss)                                  |
|   * Professional printing quality                               |
|   * Used in DTP (Desktop Publishing)                            |
|   * Can contain both vector and raster                          |
|                                                                 |
|   DISADVANTAGES:                                                |
|   * Not suitable for web                                        |
|   * Requires special viewer/software                            |
|   * Larger file size than other vector formats                  |
|                                                                 |
|   USE CASE: Print publishing, professional design               |
|                                                                 |
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5.4 PIC (PICT Format)

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|                    PIC/PICT FILE FORMAT                         |
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|                                                                 |
|   FULL NAME: PICT (Picture) Format                             |
|   DEVELOPED BY: Apple                                          |
|   FILE EXTENSION: .pic, .pict                                  |
|   TYPE: Vector and Raster                                      |
|                                                                 |
|   CHARACTERISTICS:                                              |
|   * Apple Macintosh native format                              |
|   * Supports both vector and raster                             |
|   * Uses QuickDraw graphics language                            |
|   * Supports transparency                                       |
|   * RLE compression available                                   |
|                                                                 |
|   ADVANTAGES:                                                   |
|   * High quality                                                |
|   * Supports both graphic types                                 |
```

```
|   * Native Mac format                                         |
|                                                               |
|   DISADVANTAGES:                                              |
|   * Limited Windows support                                   |
|   * Not suitable for web                                      |
|   * Outdated format (deprecated)                              |
|                                                               |
|   USE CASE: Legacy Mac applications, older design work        |
|                                                               |
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```

5.5 TIFF (Tagged Image File Format)

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|                   TIFF FILE FORMAT                             |
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|                                                                |
|   FULL NAME: Tagged Image File Format                          |
|   DEVELOPED BY: Aldus (now Adobe)                              |
|   FILE EXTENSION: .tif, .tiff                                  |
|   TYPE: Raster (lossless)                                      |
|                                                                |
|   CHARACTERISTICS:                                             |
|   * Industry standard for professional use                     |
|   * Supports tags for metadata                                 |
|   * Supports multiple compression methods                      |
|   * Can store multiple images in one file                      |
|   * Supports layers                                            |
|   * Supports color spaces (RGB, CMYK, Grayscale)               |
|   * Supports transparency                                      |
|                                                                |
|   COMPRESSION OPTIONS:                                         |
|   * None (largest file)                                        |
|   * LZW (lossless, standard)                                   |
|   * ZIP (lossless, newer)                                      |
|   * JPEG (lossy, not common)                                   |
|                                                                |
|   ADVANTAGES:                                                  |
|   * High quality                                               |
|   * Excellent metadata support                                 |
|   * Professional standard                                      |
|   * Lossless compression option                                |
|                                                                |
|   DISADVANTAGES:                                               |
|   * Large file size                                            |
|   * Not suitable for web                                       |
|   * Complex format                                             |
|                                                                |
|   USE CASE: Professional photography, medical imaging, publishing |
|                                                                |
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```

Image Format Comparison Table

```
+--------------------------------------------------------------+
|                    IMAGE FORMATS COMPARISON                  |
+--------------------------------------------------------------+
+----------+----------+----------+----------+----------+-----------+
| FORMAT   | TYPE     | COMPRESS | QUALITY  | SIZE     | BEST USE  |
+----------+----------+----------+----------+----------+-----------+
| BMP      | Raster   | None     | High     |Very Large| Windows   |
| DIB      | Raster   | None     | High     |Very Large| Apps      |
| EPS      | Vector   | Depends  | High     | Medium   | Print     |
| PIC/PICT | Both     | RLE      | High     | Medium   | Mac Legacy |
| TIFF     | Raster   | Lossless |Excellent| Large    | Professional|
| JPEG     | Raster   | Lossy    | Good     | Small    | Photos,Web |
```

```
| PNG      | Raster   | Lossless |Excellent| Medium   | Web, Logos |
| GIF      | Raster   | Lossless | Limited | Small    | Animations |
| SVG      | Vector   | --       | Excellent| Small    | Web, Icons |
| RAW      | Raster   | None     | Excellent|Very Large| Pro Photo  |
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```

6. Basics of Animation

Definition

Animation is the process of creating the illusion of motion by displaying a sequence of static images (frames) in rapid succession.

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+----------------------------------------------------------------+
|                      BASICS OF ANIMATION                       |
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```

Animation Techniques

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+----------------------------------------------------------------+
|                     ANIMATION TECHNIQUES                       |
+----------------------------------------------------------------+
|                                                                 |
|   1. TRADITIONAL ANIMATION (2D)                                 |
|      * Hand-drawn frames                                       |
|      * 24 fps for smooth motion                                |
|      * Examples: Disney classics                               |
|                                                                 |
|   2. STOP MOTION ANIMATION                                      |
|      * Physical objects photographed                           |
|      * Moved slightly between frames                           |
|      * Examples: Claymation, puppets                           |
|                                                                 |
|   3. COMPUTER ANIMATION (3D)                                    |
|      * Uses 3D modeling software                               |
|      * Rendered from 3D scenes                                 |
|      * Examples: Pixar, DreamWorks                             |
|                                                                 |
|   4. MOTION GRAPHICS                                            |
|      * Graphic design + animation                               |
|      * Titles, logos, infographics                              |
|      * Used in advertisements, presentations                    |
|                                                                 |
|   5. WEB ANIMATION                                              |
|      * GIF animations                                           |
|      * CSS animations                                           |
|      * JavaScript animations                                    |
|      * WebGL/Canvas animations                                  |
|                                                                 |
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```

12 Principles of Animation

#	Principle	Description
1	**Squash & Stretch**	Objects deform during action for natural feel
2	**Anticipation**	Prepare audience for upcoming action
3	**Staging**	Present idea clearly, focus attention
4	**Straight Ahead vs. Pose to Pose**	Frame-by-frame vs. pose-to-pose animation
5	**Follow-Through & Overlapping**	Actions continue after main motion
6	**Slow In & Slow Out**	Objects accelerate/decelerate naturally
7	**Arcs**	Natural movement follows curved paths
8	**Secondary Action**	Supporting actions enhance main action
9	**Timing**	Speed and duration of actions
10	**Exaggeration**	Emphasize actions for dramatic effect
11	**Solid Drawing**	Forms have weight, volume, and dimension
12	**Appeal**	Characters are interesting and engaging

7. Software Tools for Animation

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|                    ANIMATION SOFTWARE TOOLS                          |
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```

Software	Type	Platform	Complexity	Best For
Adobe Animate	2D	Win/Mac	Medium	Web animations, cartoons
Toon Boom	2D	Win/Mac	High	Professional 2D films
Maya	3D	Win/Mac/Linux	High	Film, TV, games
3ds Max	3D	Win	High	Games, architecture
Cinema 4D	3D	Win/Mac	Medium	Motion graphics
Blender	3D	Win/Mac/Linux	Medium	All-purpose (free)
After Effects	Motion GFX	Win/Mac	Medium	Motion graphics, VFX
Pencil2D	2D	Win/Mac/Linux	Low	Beginners, simple 2D

Visual Learning: Flowcharts

Flowchart: Image Capture Workflow

```
+-----------------------------------------------------------------------+
|                       IMAGE CAPTURE WORKFLOW                          |
+-----------------------------------------------------------------------+
+-------------------+  +-------------------+  +-------------------+
|   PHYSICAL        |  |    DIGITAL        |  |    CREATED        |
|   SOURCE          |  |    SOURCE         |  |   DIGITALLY       |
+-------------------+  +-------------------+  +-------------------+
|  * Photos         |  | * Digital         |  | * Software        |
|  * Documents      |  |   Camera          |  |   Creation        |
|  * Artwork        |  | * Smartphone      |  | * Scanning        |
|  * Film/Negative  |  | * Webcam          |  | * Tablet          |
+-------------------+  +-------------------+  +-------------------+
+-------------------+  +-------------------+  +-------------------+
|    SCANNING       |  |     CAPTURE       |  |    DIGITAL        |
|                   |  |                   |  |    CREATION       |
+-------------------+  +-------------------+  +-------------------+
| * Flatbed         |  | * Point & Shoot |  | * Photoshop       |
| * Sheetfed        |  | * DSLR            |  | * Illustrator     |
| * Drum            |  | * Mirrorless      |  | * CorelDRAW       |
| * Handheld        |  | * Action Cam      |  | * GIMP            |
+-------------------+  +-------------------+  +-------------------+
```

Flowchart: Animation Production Pipeline

```
+-----------------------------------------------------------------------+
|                     ANIMATION PRODUCTION PIPELINE                     |
+-----------------------------------------------------------------------+
+-----------------------------------------------------------------------+
|                         ANIMATION TIMELINE                            |
+-----------------------------------------------------------------------+
|                                                                        |
|   Frame 1     Frame 2     Frame 3     ...   Frame 24 (1 second)       |
|   +-------+   +-------+   +-------+         +-------+                |
|   |  O    |   |  O    |   |  O    |         |  O    |                |
|   | /|\   |   | /|\   |   | /|\   |         | /|\   |                |
|   | / \   |   | / \   |   | / \   |         | / \   |                |
|   +-------+   +-------+   +-------+         +-------+                |
|                                                                        |
|   Motion Path (Smooth Transition)                                     |
|   Keyframe ----------> Keyframe ----------> Keyframe                 |
|   (Start Position)    (Mid Position)      (End Position)              |
|                                                                        |
+-----------------------------------------------------------------------+
```

Flowchart: Image Format Selection Guide

```
+----------------------------------------------------------------------+
|                IMAGE FORMAT SELECTION GUIDE                          |
+----------------------------------------------------------------------+
+----------------------------------------------------------------------+
| FORMAT RECOMMENDATIONS                                               |
+----------------------------------------------------------------------+
|                                                                      |
| USE CASE                    RECOMMENDED FORMAT                       |
|---------------------------+------------------------------------------|
| Web Photos                | JPEG (60-80% quality)                    |
| Web Graphics/Icons        | PNG (transparent), SVG (scalable)        |
| Animations                | GIF (simple), WebP (modern)              |
| Professional Print        | TIFF (lossless), EPS (vector)            |
| Photography Archive       | RAW (original), TIFF (processed)         |
| Logo Design               | SVG, EPS, AI                             |
| Email Attachments         | JPEG, PNG (small size)                   |
| Document Scanning         | TIFF (archive), PDF (sharing)            |
| 3D Textures               | PNG (with alpha), JPEG (no alpha)        |
|                                                                      |
+----------------------------------------------------------------------+
```

Quick Reference Summary

Key Terms

1. **Raster Graphics** - Pixel-based images
2. **Vector Graphics** - Mathematical path-based images
3. **DPI (Dots Per Inch)** - Print resolution
4. **PPI (Pixels Per Inch)** - Screen resolution
5. **Bit Depth** - Number of bits per pixel
6. **CCD** - Charge-Coupled Device (camera sensor)
7. **CMOS** - Complementary Metal-Oxide-Semiconductor
8. **Lossy Compression** - Removes data, loses quality
9. **Lossless Compression** - Preserves quality
0. **Keyframe** - Important position in animation
1. **Tweening** - Creating intermediate frames
2. **FPS (Frames Per Second)** - Animation speed

Image Formats Quick Reference

```
Uncompressed/Lossless: BMP, DIB, TIFF, PNG, RAW
Vector:                EPS, SVG, AI, CDR, WMF
Compressed (Lossy):    JPEG, WebP
Animation:             GIF, APNG, WebP
```

Resolution Quick Guide

```
Web Display:    72 PPI
Email:          72-96 PPI
Photo Print:    300 DPI
Magazine:       300 DPI
```

```
Newspaper:      150-200 DPI
Billboard:      30-50 DPI
```

Image Size Formula

File Size = Width x Height x Bit Depth / 8 (bytes)

Questions

Q1: Explain the importance of graphics in multimedia with suitable examples.

Q2: Differentiate between vector and raster graphics. Explain their characteristics and use cases.

Q3: Describe various image capturing methods including scanners and digital cameras.

Q4: Explain the various attributes of images - image size, color depth, and resolution.

Q5: Calculate the file size for a 1920x1080 image with 24-bit color depth.

Q6: Compare BMP, DIB, EPS, PIC, and TIFF image file formats in detail.

Q7: What is animation? Explain the different animation techniques.

Q8: List and explain the 12 principles of animation.

Q9: Compare various animation software tools available in the industry.

Q10: Describe the complete graphics workflow in multimedia applications.

Note: These notes cover Unit III - Graphics in Multimedia as per HPU BCA Semester VI curriculum. Review all topics thoroughly for examinations.

— End of Unit III Notes —